

Files (open file)

File mode

Text	Binary	Description	Notes
'r'	'rb'	Read only	File must exist, otherwise Python raises I/O errors
'w'	'wb'	Write only	If file not exists, file is created. If file exists, Python will truncate existing data and overwrite the file.
'a'	'ab'	Append	File is in write mode only, new data will be added to the end of existing data i.e. no overwriting. If file not exists it is created
'r+'	'r+b' or 'rb+'	Read and write	File must exist, otherwise error is raised. Both reading and writing can take place
w+	'w+b' or 'wb+'	Write and read	File is created if not exists, if exists data will be truncated, both read and write allowed
'a+'	'a+b' or 'ab+'	Write and read	Same as above but previous content will be retained and both read and write.

Files (open and close file)

Examples

```
g=open("C:\\ali\\lecture\\a2.txt")  
  
h=open("C:\\ali\\lecture\\a3.txt",'w')  
  
t=open("C:\\ali\\lecture\\a4.txt",'r+')  
  
f=open("C:\\ali\\lecture\\a1.txt",'r+',encoding='utf-16')  
# any processing here  
g.close()  
h.close()  
t.close()  
f.close()
```

Dr. Ahmad

Files (Reading from File)

To read from file, python provide many functions like :

Filehandle.read([n]) : reads and return n bytes, if n is not specified it reads entire file.

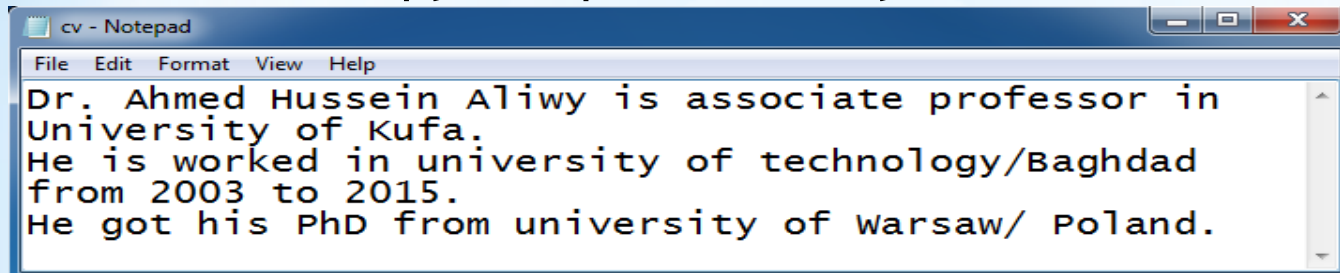
Filehandle.readline([n]) : reads a line of input. If n is specified reads at most n bytes. Read bytes in the form of string ending with line character or blank string if no more bytes are left for reading

Filehandle.readlines(): reads all lines and returns them in a list

Files (Reading from File)

Example 1: read()

To read from file, python provide many functions like :



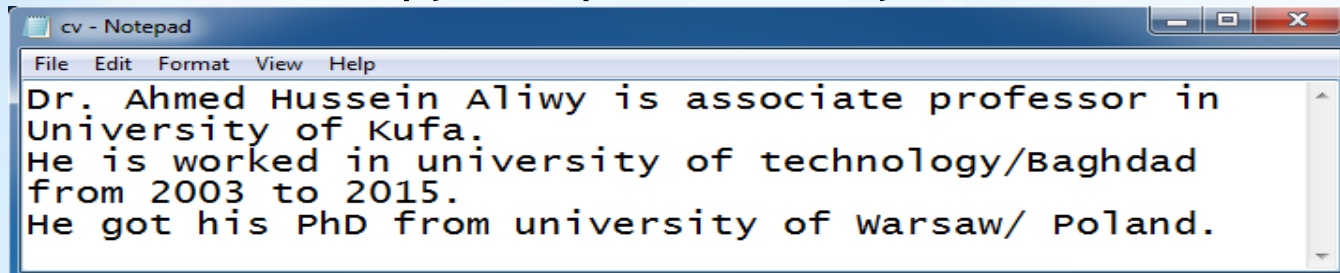
```
f=open("C:\\ali\\lecture\\cv.txt")
s1=f.read(11)
s2=f.read(5)
s3=f.read()
print("s1:", s1)
print("s2:", s2)
print("s3:", s3)
f.close()
```

```
s1: Dr. Ahmed H
s2: ussei
s3: n Aliwy is associate professor in University of
Kufa.
He is worked in university of technology/Baghdad fr
om 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

Files (Reading from File)

Example 2: read()

To read from file, python provide many functions like :



```
f=open("C:\\ali\\lecture\\cv.txt")
```

```
s1=f.read()
```

```
s2=f.read(11)
```

```
s3=f.read(5)
```

```
print("s1:", s1)
```

```
print("s2:", s2)
```

```
print("s3:", s3)
```

```
f.close()
```

```
s1: Dr. Ahmed Hussein Aliwy is associate professor  
in University of Kufa.
```

```
He is worked in university of technology/Baghdad fr  
om 2003 to 2015.
```

```
He got his PhD from university of Warsaw/ Poland.
```

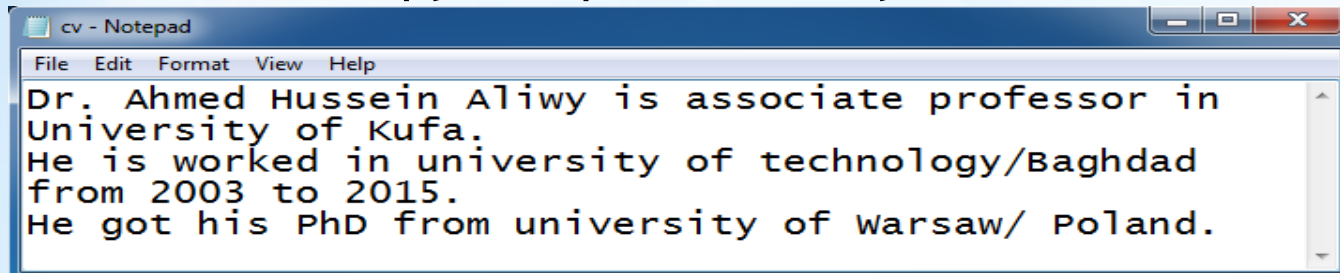
```
s2:
```

```
s3:
```

Files (Reading from File)

Example 3: readline()

To read from file, python provide many functions like :



```
f=open("C:\\ali\\lecture\\cv.txt")
L1=f.readline()
L2=f.readline()
L3=f.readline(10)
print(L1)
print(L2)
print(L3)
f.close()
```

Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.

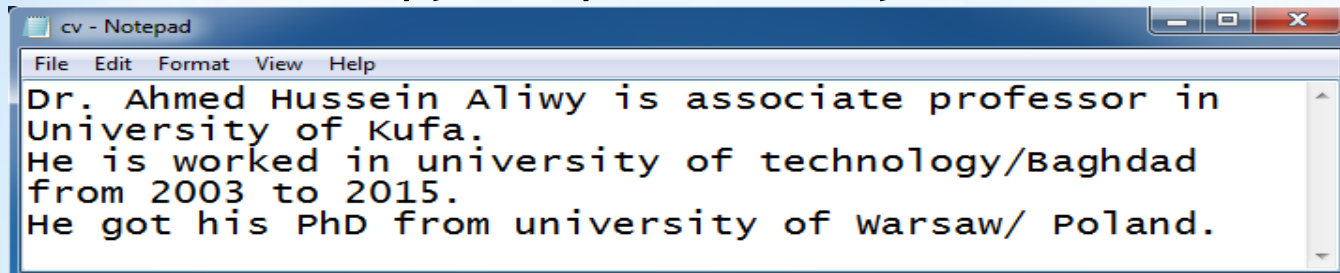
He is worked in university of technology/Baghdad
from 2003 to 2015.

He got his

Files (Reading from File)

Example 4: readline()

To read from file, python provide many functions like :

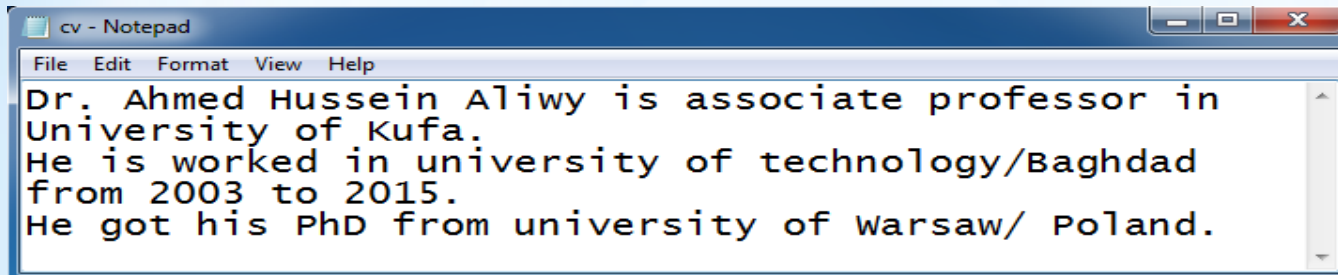


```
f=open("C:\\ali\\lecture\\cv.txt")
L1=f.readline()
L2=f.readline()
L3=f.readline(10)
print(L1,end='')
print(L2,end='')
print(L3,end='')
f.close()
```

Dr. Ahmed Hussein Aliwy is associate professor in University of Kufa.
He is worked in university of technology/Baghdad from 2003 to 2015.
He got his

Files (Reading from File)

Example 5: read files line by line



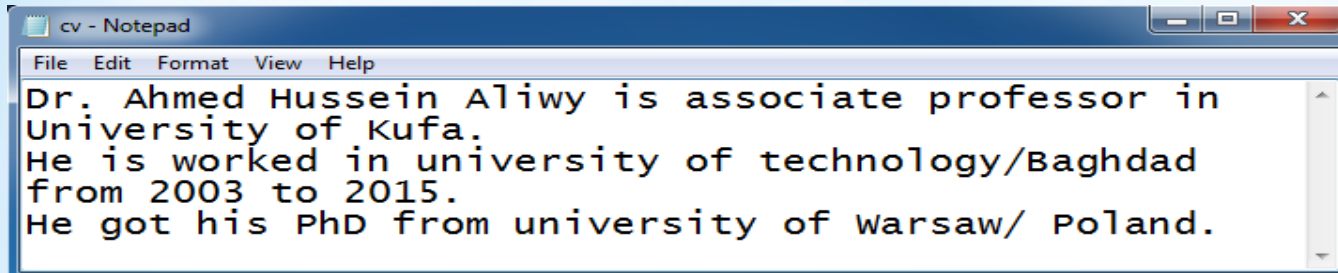
```
cv - Notepad
File Edit Format View Help
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

```
f=open("C:\\ali\\lecture\\cv.txt")
for st in f:
    print(st,end='')
```

Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.

Files (Reading from File)

Example 6: readlines()



```
f=open("C:\\ali\\lecture\\cv.txt")
content=f.readlines()
print(content)
print("first line is:",content[0])
print("second line is:",content[1])
['Dr. Ahmed Hussein Aliwy is associate professor
in University of Kufa.\n', 'He is worked in unive
rsity of technology/Baghdad from 2003 to 2015.\n'
, 'He got his PhD from university of Warsaw/ Pola
nd.']
first line is: Dr. Ahmed Hussein Aliwy is associa
te professor in University of Kufa.

second line is: He is worked in university of tec
hnology/Baghdad from 2003 to 2015.
```

Files (writing to File)

To write to file, python provide many functions like :

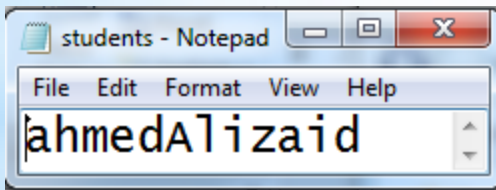
Name	Syntax	Description
<code>write()</code>	<code>Filehandle.write(S1)</code>	Writes string S1 to file referenced by filehandle
<code>Writelines()</code>	<code>Filehandle.writelines(L)</code>	Writes all string in List L as lines to file referenced by filehandle.

Files (Writing to File)

Example-1: write() using “w” mode

```
f=open("C:\\ali\\lecture\\students.txt", 'w')
for i in range(3):
    name=input('Enter the name: ')
    f.write(name)
f.close()
```

```
Enter the name:  ahmed
Enter the name:  Ali
Enter the name:  zaid
```

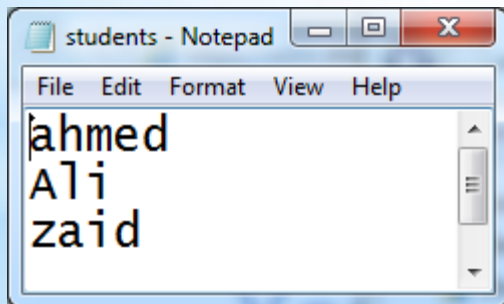


Files (Writing to File)

Example-1: write() using “w” mode

```
f=open("C:\\ali\\lecture\\students.txt", 'w')
for i in range(3):
    name=input('Enter the name: ')
    f.write(name)
    f.write('\n')
f.close()
```

```
Enter the name: ahmed
Enter the name: Ali
Enter the name: zaid
```

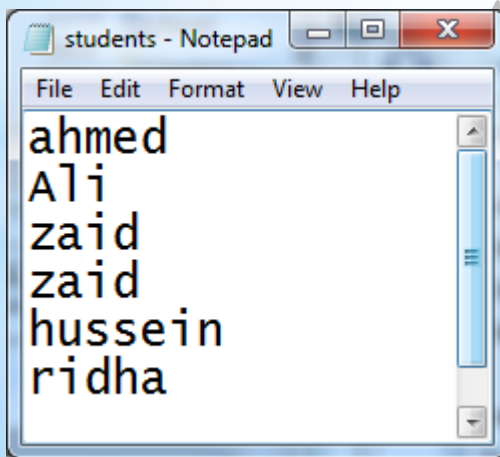


Files (Writing to File)

Example-1: write() using "a" mode

```
f=open("C:\\ali\\lecture\\students.txt", 'a')
for i in range(3):
    name=input('Enter the name: ')
    f.write(name)
    f.write('\n')
f.close()
```

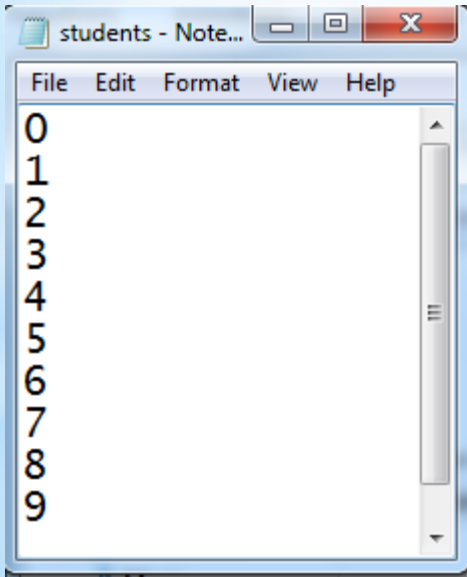
```
Enter the name:  zaid
Enter the name:  hussein
Enter the name:  ridha
```



Files (Writing to File)

Example-1: writelines () using “w” mode

```
f=open("C:\\ali\\lecture\\students.txt", 'w')
L=[]
for i in range(10):
    L.append(str(i)+'\n')
f.writelines(L)
f.close()
```



Files (exercise)

If you have a text file called "history.txt" , do the following separately:

1- Write a function in python to read the content from a text file "history.txt" line by line and display the same on screen.

```
def read_file():  
    file = open("history.txt", "r")  
    for line in file:  
        print(line, end="")  
    file.close()
```

read_file()

2- Write a function in python to count the number of lines from a text file "story.txt" which is not starting with an alphabet "A"

3- Write a function in Python to count and display the total number of words in a text file.

```
def count_words():  
    file = open("history.txt", "r")  
    count = 0  
    for line in file:  
        words = line.split()  
        count += len(words)  
    print("Total words are", count)
```

count_words()

Files (exercise)

If you have a text file called "history.txt" , do the following separately:

- 4- Write a function `display_words()` in python to read lines from the text file , and display those words, which are less than 4 characters.
- 5- Write a function in Python to count the words "and" and "or" present in the text file . [Note that the words "and" and "or" are complete words]
- 6- Write a function in Python to count words in a text file those are ending with alphabet "f".
- 7- Write a function in Python to count uppercase character in a text file.
- 8- A text file named "matter.txt" contains some text, which needs to be displayed such that every next character is separated by a symbol "#". Write a function definition for `hash_display()` in Python that would display the entire content of the file in the desired format.

Example :

If the file `history.txt` has the following content stored in it :

THE WORLD IS ROUND

The function `hash_display()` should display the following content :

T#H#E# #W#O#R#L#D# #I#S# #R#O#U#N#D#

Files (exercise)

- 9- Write a function `display_words()` in python to read lines from the text file , and display those words, which are less than 4 characters.
- 5- write python program to create new file “myfile.txt” and write the following lines in it:
 - my name is Ahmed
 - I have a car and bicycle
- 6- write Python program to write the line “ ahmed hussein” to the end of this file:
- 7- write Python program to read the content of one file line by line and stor it in another file line by line.

Exception handling

Dr. Ahmed Hussein Aliwy